

Q Will my baby's brain be fully formed when she is born or does it continue growing?

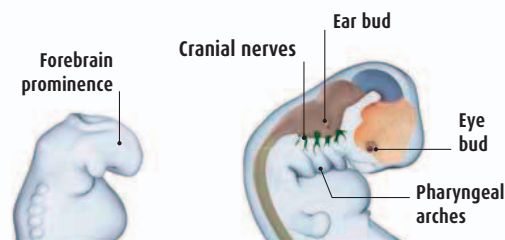
Your baby's brain grows faster than any other organ and, consequently, the head is still out of proportion with the rest of the body when she is born.

The brain continues growing and developing throughout childhood and into adulthood, but is at its most sensitive during certain periods in the early years. Between conception and about the age of three a young child's brain goes through an enormous amount of change. At birth, your baby will have around 100 billion brain cells (neurons) and that's roughly the number she has throughout her life. At this time, basic neural connections are in place, which help to control vital functions such as breathing, heart rate, digestion, and reflexes. The connections are formed at a very fast rate during the first few years and, as more links form, higher mental functions such as memory, increased attention span, language, intellect, and social skills develop. By adulthood, the neural network allows for reasoning, judgment, and original thought.

BRAIN FORMATION

During her time in the uterus, your growing baby's brain goes through an extraordinary amount of growth and development.

WEEK	ACTIVITY
5	The neural tube divides to create separate spaces for the development of her brain cells and nerve cells—effectively beginning the formation of her brain and central nervous system.
7	The two hemispheres of your baby's brain begin to separate. The right one will go on to govern her creativity, spatial awareness, and lateral thinking; the left will influence her logic and practical thinking.
8	The synapses, which open and close the pathways to send messages between neurons, start to form. Your baby is born with relatively few synapses; most of them form within the first two to three years of life as memories and her understanding of the world are laid down.
12	Your baby's cerebellum begins to form. At the same time, the two halves of her brain begin to communicate with each other and there's an enormous burst of neuron formation (called neurogenesis). Her hypothalamus and pituitary glands have formed.
14-16	Reflex responses in your baby's limbs and facial features suggest that she's now responding to stimuli in the uterus. This confirms that her brain's neurons are beginning to work, processing what she is experiencing. This coincides with maturing taste buds and the ability of the bones in her ear to transmit sound.
22-25	The formation of her brain matter is complete and her cerebral cortex starts to fold over on itself, creating her brain's typical ridged, wrinkled appearance. This increases her brain's surface area, which optimizes her chances for learning.
26-40	Your baby's nervous system is now fully developed. Her cerebral cortex begins to send out electrical impulses: researchers believe that memory has begun to develop because some of these impulses respond to the sound of familiar voices. Her cerebellum is three times bigger at 40 weeks than it was at 22 weeks, and her whole brain weighs approximately 12 oz (350 g).



At 5 weeks The neural tube forms the forebrain prominence.

At 9 weeks Swellings that will become different parts of the brain are growing and are beginning to fold into one another.

Cerebrum The largest part of the brain and is divided into the left and right hemispheres

Cerebellum The region of the brain that is key to coordination, attention, and language

Pons

Spinal cord

